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Remote explosive separation tool

Abstract

Exemplary embodiments of apparatus and methods for remotely and explosively cutting the casing of a UXO and separating the fuzing components from the main charge in the UXO. A remote explosive separation tool creates a Linear Explosively Formed Projectile (LEFP) charge to separate fuzing components from live explosive threats in UXO, for example only, a 155 mm artillery round. The tool functions by explosively deforming a copper liner into a linear projectile or penetrator which then impacts the UXO casing. The speed of impact enables the fuzing elements of the UXO to be removed from the main charge of the UXO either by directly cutting the fuze elements off or by imparting the fuze elements with enough momentum such that they break off. The process of dismembering the fuze elements occurs without detonating the main charge of the UXO. The tool may be remotely placed by an unmanned vehicle.

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Background/Summary

BACKGROUND OF THE INVENTION

(1) UXO may be rendered safe by separating the fuzing components from the main charge in the UXO. FIG. 1 is a partially cutaway schematic of an embodiment of an unexploded 155 mm projectile 100. Projectile 100 may include a base 110, a steel casing 120, an explosive charge 130, fuze components 140, and a fuze well having a bottom 150. If the fuze components 140 are separated from the main charge 130, the separated fuze components may explosively function or still be unexploded and not safe to handle. However, the separated fuze components 140 cannot set off the main charge 130 and cause further hazard. For UXO projectiles such as the 155 mm explosive projectile 100, it is necessary to penetrate or cut into the steel casing 120 of the projectile below the bottom 150 of the fuze well to separate, effectively, the fuze components 140 from the main charge 130.

(2) Conventional technologies may use manually placed charges for penetrating the steel casing of a projectile. Manual placement of these penetrating charges may be very dangerous because of the risk of the UXO exploding and injuring or killing any nearby humans.

(3) A need exists for apparatus and methods to remotely and securely cut through a UXO projectile casing and separate the fuzing components from the main charge in the UXO.

SUMMARY OF THE INVENTION

(4) One aspect of the invention is a remote explosive separation tool for forming a Linear Explosively Formed Projectile (LEFP). The remote explosion separation tool may include a housing having an open front, two opposing side walls, a rear wall, and top and bottom walls. A quantity of C-4 explosive may be packed in the housing. A linear concave copper liner may be disposed in the housing adjacent to the quantity of C-4 explosive. The linear concave copper liner may face the open front of the housing and have an elliptical cross-section. A pair of parallel stand-off wands may extend from respective opposing side walls of the housing. The pair of parallel stand-off wands may include, at distal ends, visual indicators for identifying a point of impact of the LEFP. A base may be attached to the bottom wall of the housing. The base may be configured to contact a ground surface and stably support the tool so that the tool does not tip over prior to detonation of the quantity of C-4 explosive. An unmanned vehicle interface may be attached to the top wall of the housing for attaching the tool to an unmanned vehicle grip.

(5) The remote explosive separation tool may include a pair of wand holder sleeves located on the respective opposing side walls of the housing. Proximal ends of the pair of parallel stand-off wands may be disposed in respective ones of the pair of wand holder sleeves. A proximal end of each stand-off wand may include a barb that contacts the rear surface of the housing.

(6) The base of the remote explosive separation tool may be one of a flat static stand, an angled static stand, and an articulating stand. The unmanned vehicle interface may include an unmanned vehicle grip for one of an inline ground robot, a sideways ground robot, and an unmanned aerial vehicle.

(7) In another aspect, a method of separating fuzing components from a main charge in unexploded ordnance (UXO) may include providing the remote explosive separation tool of the first aspect of the invention. A Linear Explosively Formed Projectile (LEFP) may be created from the linear concave copper liner having the elliptical cross-section. The LEFP may be directed at the UXO. Using the LEFP, the fuzing components may be separated from the main charge without setting off or detonating the main charge.

(8) The method may include transporting the remote explosive separation tool to a location near the UXO, using an unmanned vehicle. The method may include using the unmanned vehicle to aim the remote explosive separation tool at the UXO. The UXO may have a steel casing that is penetrated

by the Linear Explosively Formed Projectile (LEFP).

(9) The invention will be better understood, and further objects, features, and advantages thereof will become more apparent from the following description of the preferred embodiments, taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings, which are not necessarily to scale, like or corresponding parts are denoted by like or corresponding reference numerals.

(2) FIG. 1 is a schematic, partially in section, of an embodiment of a conventional 155 mm projectile.

(3) FIG. 2 is a perspective view of an exemplary embodiment of a remote explosive separation tool.

(4) FIG. 3 is transverse cutaway view, partially in section, through a portion of the tool of FIG. 2.

(5) FIG. 4 is a perspective view of the tool of FIG. 2 with an exemplary embodiment of a flat static base.

(6) FIG. 5 is a perspective view of the flat static base of FIG. 4.

(7) FIG. 6 is a perspective view of the tool of FIG. 2 with an exemplary embodiment of an angled static base.

(8) FIG. 7 is a perspective view of the angled static base of FIG. 6.

(9) FIG. 8 is a perspective view of the tool of FIG. 2 with an exemplary embodiment of an articulating base.

(10) FIG. 9A is a perspective view of the bottom portion of the base of FIG. 8.

(11) FIG. 9B is a perspective view of the top portion of the base of FIG. 8.

(12) FIG. 10 is a partial perspective view of the tool of FIG. 2 with an exemplary embodiment of an inline ground robot grip.

(13) FIG. 11 is a partial perspective view of the tool of FIG. 2 with an exemplary embodiment of a sideways ground robot grip.

(14) FIG. 12 is a partial perspective view of the tool of FIG. 2 with an exemplary embodiment of an unmanned aerial vehicle grip.

(15) FIG. 13 is a perspective view of the tool of FIG. 2 in position to fire a linear explosively formed projectile at an unexploded 155 mm projectile.

(16) FIG. 14 is another perspective view of FIG. 13.

DETAILED DESCRIPTION OF THE INVENTION

(17) Disclosed herein are exemplary embodiments of apparatus and methods for remotely and explosively cutting the casing of a UXO and separating the fuzing components from the main charge in the UXO. A remote explosive separation tool creates a Linear Explosively Formed Projectile (LEFP) that may separate fuzing components from live explosive threats in UXO, for example, a 155 mm artillery round. The remote explosive separation tool functions by explosively deforming a copper liner into a linear projectile or penetrator (as opposed to a jet from a shaped charge) which then impacts the UXO casing at a speed in the range of about 5000 feet per second to about 7000 feet per second. In an exemplary embodiment, the impact speed of the LEFP is about 6000 feet per second. The speed of impact enables the fuzing elements of the UXO to be removed from the main charge of the UXO either by directly cutting the fuze elements off or by imparting the fuze elements with enough momentum such that they break off. The process of dismembering the fuze elements occurs without setting off or detonating the main charge of the UXO. The tool may be remotely placed by, for example, a ground robot or an unmanned aerial vehicle.

(18) FIG. 2 is a perspective view of one embodiment of a remote explosive separation tool **10** (hereinafter “tool”). FIG. 3 is transverse cutaway view through a portion of the tool **10** of FIG. 2.

The tool **10** may include a housing **12** in the shape of a rectangular prism. The housing **12** may be made of, for example, plastic. The housing **12** may be printed by a 3D printer. The housing **12** may have an open front **14**; two opposing side walls **16, 18**; a rear wall **20**; and top and bottom walls **22, 24**. A detonator well **26** for holding a detonator (not shown) may be located on the housing **12**, for example, on the rear wall **20**. The detonator may be initiated in a variety of known ways, including wirelessly. A quantity of C-4 explosive **28** may be packed in the housing **12**. In one embodiment, the quantity of C-4 explosive **28** may be, for example, about 1.25 pounds. Other embodiments may use other quantities of C-4 explosive **28**. The C-4 explosive **28** may extend from the top wall **22** to the bottom wall **24**. A linear concave copper liner **30** may be disposed adjacent to the quantity of C-4 explosive **28**. The linear concave copper liner **30** may extend from the top wall **22** to the bottom wall **24**. The linear concave copper liner **30** may face the open front **14** of the housing **12**. The linear concave copper liner **30** may have an elliptical profile or cross-section **32**, as opposed to the circular profile commonly used in explosively formed projectiles. The LEFP forms a solid “blade-like” projectile that experiences minimal break-up at stand-off, thereby allowing the tool **10** to be used at long stand-offs (at least 4 charge diameters) compared with a linear shaped charge (1 charge diameter).

(19) A pair of parallel stand-off wands **34** may extend from respective opposing side walls **16, 18** between the top and bottom walls **22, 24**. The wands **34** may extend from the midpoints of the opposing side walls **16, 18**. A pair of wand holder sleeves **36** may be located on respective opposing side walls **16, 18**. Proximal ends of the pair of parallel stand-off wands **34** may be disposed in respective ones of the pair of wand holder sleeves **36**. The pair of parallel stand-off wands **34** may extend perpendicular to the plane of the open front **14**. Barbs **38** at the proximal ends of the wands **34** may bear against the rear wall **20** of the housing **12** to help maintain the position of the wands. Distal ends of the wands **34** may include visual indicators **40** for identifying a point of impact of the LEFP and for help in remote placement of the tool **10** by an unmanned vehicle. The visual indicators **40** may include whiskers **42** and a reticle **44**. The reticle **44** may indicate the central point of impact of the LEFP, which may correspond to a projection of the geometric center of the open front **14** of the housing **12** to a point that is midway between the distal ends of the stand-off wands **34**. The stand-off wands **34** enable the tool **10** be placed at the optimal distance from a target, without measurement, at the time of remotely placing the tool. The optimal distance may vary depending on the target so wands **34** having different lengths may be used.

(20) A base interface **39** (FIG. 2) may be located on the bottom wall **24** of the housing **12**. The base interface **39** may include one part of a dovetail joint. The base interface **39** may be joined to one of several types of bases **46, 48** or **50** (see FIGS. 4, 6, and 8, respectively). Bases **46, 48**, and **50** share some features with the stand disclosed in U.S. Pat. No. 11,572,976 issued on Feb. 7, 2023 and entitled “Multiple Angle Pivoting Placement (MAPP) Stand.” The entirety of U.S. Pat. No. 11,572,976 is expressly incorporated by reference herein. As seen in FIGS. 4 and 5, base **46** is a flat static base and includes on its top surface a part **47** of a dovetail joint for mating with the dovetail joint on base interface **39** (FIG. 2). As seen in FIGS. 6 and 7, base **48** is an angled static base and includes on its top surface a part **49** of a dovetail joint for mating with the dovetail joint on base interface **39**. The dovetail joint part **49** is angled so that housing **12** will be at a fixed angle with respect to the bottom surface of the base **48**. The angle of dovetail joint part **49** may be varied by manufacturing different bases **48** using, for example, 3D printing. As seen in FIGS. 8, 9A and 9B, base **50** is an articulating base and includes on its top surface a part **51** of a dovetail joint for mating with the dovetail joint on base interface **39**. Referring to FIGS. 9A and 9B, base **50** includes a ground-contacting bottom portion **52**, a socket **54**, and a ball **56**. Socket **54** receives the ball **56** shown in FIG. 9B to form a friction fit so that the ball **56** may be rotated or angled in the socket **54** as needed to aim the tool **10**. The ball **56** includes the dovetail joint portion **51** that interfaces with the dovetail joint portion on base interface **39** on the bottom wall **24** of the housing **12**. The fit between the ball **56** and socket **54** may be, for example, pre-adjusted by an optional clamp (not

shown), such as a zip tie, which is disposed around the exterior of the socket and passes through loops **58** on the socket. Optional rubber grips (not shown) may be placed in slots **60** in the socket **54**. An optional friction grip **62** may be disposed on the interior circumference of the socket **54**. The bases **46**, **48**, **50** may be configured to contact the ground surface and stably support the tool **10** so that the tool does not tip over prior to detonation of the quantity of C-4 explosive **28**.

(21) Referring to FIG. 2, an unmanned vehicle interface **41** may be attached to the top wall **22** of the housing **12** for interfacing with an unmanned vehicle grip. The unmanned vehicle interface **41** may include one part of a dovetail joint. The unmanned vehicle interface **41** may be joined to one of several types of unmanned vehicle grips. The unmanned vehicle grips may include another part of a dovetail joint for joining to the unmanned vehicle interface **41**. An unmanned vehicle attaches to the unmanned vehicle grip to transport the tool **10** to a location of use of the tool. The unmanned vehicle grips may include, for example, an inline ground robot grip **64** (FIG. 10); a sideways ground robot grip **66** (FIG. 11); and an unmanned aerial vehicle grip **68** (FIG. 12). A system of lines (not shown) may be attached to the unmanned aerial vehicle grip **68** and in turn may be attached to the unmanned aerial vehicle (not shown).

(22) FIG. 13 is a perspective view of the tool **10** of FIG. 2 in position to fire a linear explosively formed projectile at an unexploded 155 mm projectile **100**. FIG. 14 is another perspective view of FIG. 13. The tool **10** of FIGS. 13 and 14 includes the inline ground robot grip **64** and the articulating base **50**. The tool **10** has been aimed at the projectile **100** using the unmanned vehicle (not shown) that transported the tool to the area adjacent to the projectile. The visual indicator **40** of the stand-off wands **34** includes whiskers **42** and a reticle **44**. Initiation of a detonator in the detonator well **26** causes the quantity of C-4 explosive **28** to explode and explosively transform the copper liner **30** into a Linear Explosively Formed Projectile (LEFP). The LEFP penetrates the steel casing **120** of the projectile **100** and separates the fuze components **140** from the main explosive charge **130**. The process of dismembering the fuze components **140** occurs without setting off or detonating the main charge **130**. Even if the separated fuze components **140** may explosively function or still be unexploded and not safe to handle, the separated fuze components cannot set off the main charge **130** and cause further hazard.

(23) Actual Test Results

(24) First test: The remote explosive separation tool successfully cut inert fuze components from an inert 155 mm projectile.

(25) Second test: The remote explosive separation tool successfully cut inert fuze components from a live 155 mm projectile.

(26) Third test: The remote explosive separation tool successfully cut live, armed, unexploded fuze components from a live 155 mm projectile containing IMX-101 as the main charge.

(27) Fourth test: The remote explosive separation tool successfully cut live, armed, unexploded fuze components from a live 155 mm projectile containing TNT as the main charge.

(28) Finally, any numerical parameters set forth in the specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the present invention. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of significant digits and by applying ordinary rounding.

Claims

1. A remote explosive separation tool for forming a Linear Explosively Formed Projectile (LEFP), comprising: a housing in the shape of a rectangular prism, the housing includes an open front, two opposing side walls, a rear wall, and top and bottom walls; a detonator well being located on the housing; a quantity of C-4 explosive being packed in the housing and extending from the top wall to the bottom wall; a linear concave copper liner being disposed adjacent to the quantity of C-4

explosive and extending from the top wall to the bottom wall, the linear concave copper liner faces the open front of the housing and includes an elliptical cross-section; a pair of parallel stand-off wands extending from respective opposing side walls midway between the top and bottom walls, the pair of parallel stand-off wands extends perpendicular to a plane of the open front and includes, at distal ends, visual indicators for identifying a point of impact of the LEFP; a base interface being attached to the bottom wall of the housing for attaching the housing to a base, the base being configured to contact a ground surface and stably support the tool so that the tool does not tip over prior to detonation of the quantity of C-4 explosive; and an unmanned vehicle interface being attached to the top wall of the housing for attaching the tool to an unmanned vehicle grip.

2. The remote explosive separation tool of claim 1, wherein the point of impact of the LEFP corresponds to a projection of a geometric center of the open front of the housing to a second point that is midway between the distal ends of the stand-off wands.
3. The remote explosive separation tool of claim 1, wherein the housing is a plastic housing.
4. The remote explosive separation tool of claim 1, wherein the base interface includes one part of a dovetail joint and the base includes another part of a dovetail joint.
5. The remote explosive separation tool of claim 1, wherein the quantity of C-4 explosive is about 1.25 pounds.
6. The remote explosive separation tool of claim 4, wherein the unmanned vehicle interface includes one part of a dovetail joint and the unmanned vehicle grip includes another part of the dovetail joint.
7. The remote explosive separation tool of claim 1, wherein the unmanned vehicle grip is one of an inline ground robot grip, a sideways ground robot grip, and an unmanned aerial vehicle grip.
8. The remote explosive separation tool of claim 1, wherein the base is one of a flat static stand, an angled static stand, and an articulating stand.
9. The remote explosive separation tool of claim 1, further comprising a pair of wand holder sleeves being located on respective opposing side walls of the housing, wherein proximal ends of the pair of parallel stand-off wands are disposed in respective ones of the pair of wand holder sleeves.
10. The remote explosive separation tool of claim 9, wherein the visual indicators at the distal ends of the pair of parallel stand-off wands include whiskers and a bullseye.
11. A method of separating fuzing components from a main charge in unexploded ordnance (UXO), comprising: providing the remote explosive separation tool of claim 1; creating a Linear Explosively Formed Projectile (LEFP) from the linear concave copper liner having the elliptical cross-section; and directing the LEFP at the UXO and separating the fuzing components from the main charge without at least one of setting off and detonating the main charge.
12. The method of claim 11, further comprising transporting the remote explosive separation tool to a location near the UXO, using an unmanned vehicle.
13. The method of claim 12, further comprising, using the unmanned vehicle, aiming the remote explosive separation tool at the UXO.
14. The method of claim 12, further comprising, using the unmanned vehicle, aiming the remote explosive separation tool at the UXO, wherein the UXO includes a steel casing, and wherein the Linear Explosively Formed Projectile (LEFP) penetrates the steel casing.
15. The method of claim 12, further comprising, using the unmanned vehicle, aiming the remote explosive separation tool at the UXO, wherein the UXO includes a steel casing, wherein the Linear Explosively Formed Projectile (LEFP) penetrates the steel casing, wherein the UXO is a 155 mm projectile.
16. A remote explosive separation tool for forming a Linear Explosively Formed Projectile (LEFP), comprising: a housing including an open front, two opposing side walls, a rear wall, and top and bottom walls; a quantity of C-4 explosive being packed in the housing; a linear concave copper liner being disposed adjacent to the quantity of C-4 explosive, the linear concave copper liner faces

the open front of the housing and includes an elliptical cross-section; a pair of parallel stand-off wands extending from respective opposing side walls, the pair of parallel stand-off wands includes at distal ends, visual indicators to identify a point of impact of the LEFP; a base being attached to the bottom wall of the housing, the base is configured to contact a ground surface and stably support the tool so that the tool does not tip over prior to detonation of the quantity of C-4 explosive; and an unmanned vehicle interface being attached to the top wall of the housing for attaching the tool to an unmanned vehicle grip.

17. The remote explosive separation tool of claim 16, further comprising a pair of wand holder sleeves being located on the respective opposing side walls of the housing, wherein proximal ends of the pair of parallel stand-off wands are disposed in respective ones of the pair of wand holder sleeves.

18. The remote explosive separation tool of claim 17, wherein a proximal end of each stand off wand includes a barb that contacts the rear surface of the housing.

19. The remote explosive separation tool of claim 16, wherein the base is one of a flat static stand, an angled static stand, and an articulating stand.

20. The remote explosive separation tool of claim 16, wherein the unmanned vehicle interface includes an unmanned vehicle grip for one of an inline ground robot, a sideways ground robot, and an unmanned aerial vehicle.
